



Configuring One-Armed Routing to Implement Inter-VLAN Routing

Difficulty (out of 3 stars): ★★★

Recommended study time: 1 hour

Contents

1 About This Lab	2
1.1 Task Description	2
1.2 Objectives	2
2 Task Preparation	3
2.1 Network Topology	3
2.2 Initial Configuration	3
3 Task Implementation	4
3.1 Checking Basic Configurations	4
3.2 On S1, assign R2 and R3 to different VLANs.	4
3.3 Configure a trunk port on S1.	4
3.4 Configure a dot1q termination sub-interface on R1.	4
3.5 Testing the Connectivity Between VLANs	5
3.6 Configuring OSPF	5
4 Verification	8
5 Quiz	8



1 About This Lab

1.1 Task Description

VLANs are separated at Layer 2 to minimize broadcast domains. To enable the communication between VLANs,

Huawei provides a variety of technologies. The following two technologies are commonly used:

- Dot1q termination sub-interface: Such sub-interfaces are Layer 3 logical interfaces. Similar to a VLANIF interface, after a dot1q termination sub-interface and its IP address are configured, the device adds the corresponding MAC address entry and sets the Layer 3 forwarding flag to implement Layer 3 communication between VLANs. A dot1q termination sub-interface applies to scenarios where a Layer 3 Ethernet interface connects to multiple VLANs.
- VLANIF interface: VLANIF interfaces are Layer 3 logical interfaces. After a VLANIF interface and its IP address are configured, the device adds the MAC address and VID of the VLANIF interface to the MAC address table and sets the Layer 3 forwarding flag of the MAC address entry. When the destination MAC address of a packet matches the entry, the packet is forwarded at Layer 3 to implement Layer 3 communication between VLANs.

In this lab, use dot1q termination sub-interfaces to implement inter-VLAN communication. After completing this task, you will be able to understand inter-VLAN communication and be familiar with the method of implementing inter-VLAN communication using dot1q termination sub-interfaces.

1.2 Objectives

On completing this task, you will be able to:

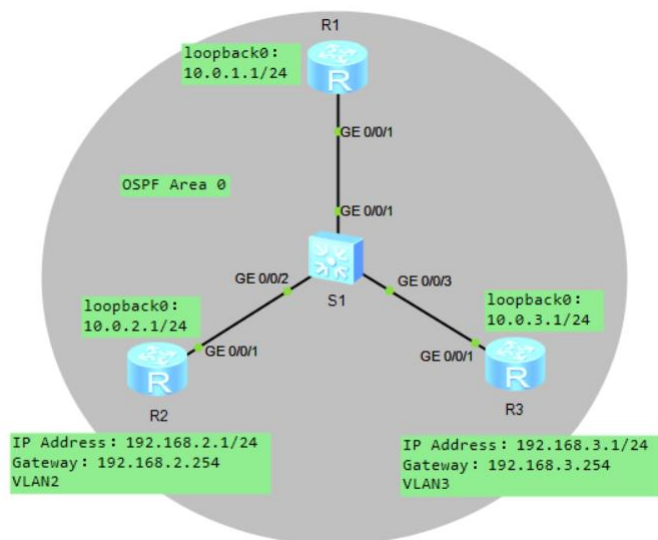
1. Learn how to use dot1q termination sub-interfaces to implement inter-VLAN communication.
2. Understand the forwarding process of inter-VLAN communication.



2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



1. Simulate terminal users on R2 and R3 and assign IP addresses 192.168.2.1/24 and 192.168.3.1/24 to the interfaces.
2. The gateway addresses of R2 and R3 are 192.168.2.254 and 192.168.3.254, respectively.
3. On S1, assign GigabitEthernet0/0/2 and GigabitEthernet0/0/3 to VLAN 2 and VLAN 3, respectively.

2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 192.168.2.1 24
[R2-GigabitEthernet0/0/1]quit
[R2]ip route-static 0.0.0.0 0.0.0.0 192.168.2.254 # Configure a default route, which is
equivalent to configuring a gateway for the device.
[R3]
```



- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int g0/0/1
[R3-GigabitEthernet0/0/1]ip address 192.168.3.1 24
[R3-GigabitEthernet0/0/1]quit
[R3]ip route-static 0.0.0.0 0.0.0.0 192.168.3.254 # Configure a default route, which is
equivalent to configuring a gateway for the device.
[R3]
```

- Initial configuration of S1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

3 Task Implementation

3.1 Checking Basic Configurations

Omitted

3.2 On S1, assign R2 and R3 to different VLANs.

```
[S1]vlan batch 2 3
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1]interface GigabitEthernet 0/0/2
[S1-GigabitEthernet0/0/2]port link-type access
[S1-GigabitEthernet0/0/2]port default vlan 2
[S1-GigabitEthernet0/0/2]quit
[S1]interface GigabitEthernet 0/0/3
[S1-GigabitEthernet0/0/3]port link-type access
[S1-GigabitEthernet0/0/3]port default vlan 3
```

3.3 Configure a trunk port on S1.

```
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]port trunk allow-pass vlan 2 3
```

The link between S1 and R1 must allow packets from VLAN 2 and VLAN 3 to pass through because R1 needs to terminate the VLAN tags of packets exchanged between VLANs.

3.4 Configure a dot1q termination sub-interface on R1.

```
[R1]interface GigabitEthernet 0/0/1.2
```

Create a sub-interface and enter the sub-interface view. In this example, **2** indicates the sub-interface ID. It is recommended that the sub-interface ID be the same as the VLAN ID.

```
[R1-GigabitEthernet0/0/1.2]dot1q termination vid 2
```



The **dot1q termination vid *vlan-id*** command configures the VLAN ID for dot1q termination on a sub-interface.

In this example, when GigabitEthernet0/0/1 receives data tagged with VLAN 2, it sends the data to sub-interface 2 for VLAN termination and subsequent processing. The data sent from sub-interface 2 is also tagged with VLAN 2.

```
[R1-GigabitEthernet0/0/1.2]arp broadcast enable
```

By default, a VLAN tag termination sub-interface discards all received broadcast packets instead of forwarding them. To allow a VLAN tag termination sub-interface to forward broadcast packets, run the **arp broadcast enable** command on the sub-interface to enable ARP broadcast. By default, this function is enabled on some devices. The command configuration varies according to the device.

```
[R1-GigabitEthernet0/0/1.2]ip address 192.168.2.254 24
[R1-GigabitEthernet0/0/1.2]quit
[R1]interface GigabitEthernet 0/0/1.3
[R1-GigabitEthernet0/0/1.3]dot1q termination vid 3
[R1-GigabitEthernet0/0/1.3]arp broadcast enable
[R1-GigabitEthernet0/0/1.3]ip address 192.168.3.254 24
[R1-GigabitEthernet0/0/1.3]quit
```

3.5 Testing the Connectivity Between VLANs

```
<R2>ping 192.168.3.1
PING 192.168.3.1: 56 data bytes, press CTRL_C to break
Reply from 192.168.3.1: bytes=56 Sequence=1 ttl=254 time=60 ms
Reply from 192.168.3.1: bytes=56 Sequence=2 ttl=254 time=40 ms
Reply from 192.168.3.1: bytes=56 Sequence=3 ttl=254 time=110 ms
Reply from 192.168.3.1: bytes=56 Sequence=4 ttl=254 time=70 ms
Reply from 192.168.3.1: bytes=56 Sequence=5 ttl=254 time=100 ms

--- 192.168.3.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 40/76/110 ms

<R2>tracert 192.168.3.1
tracert to 192.168.3.1(192.168.3.1), max hops: 30 ,packet length: 40,press CTRL_C to break

1 192.168.2.254 30 ms 50 ms 50 ms
2 192.168.3.1 70 ms 60 ms 60 ms
```

VLAN 2 and VLAN 3 can communicate with each other.

3.6 Configuring OSPF

1. Modify the configurations of R2 and R3. Delete the static routes from R2 and R3.

```
[R2]undo ip route-static 0.0.0.0 0.0.0.0 192.168.2.254
```

```
[R3]undo ip route-static 0.0.0.0 0.0.0.0 192.168.3.254
```



2. Add loopback0 interfaces on R1, R2, and R3.

```
[R1]int loopback 0
[R1-LoopBack0]ip address 10.0.1.1 24
[R1-LoopBack0]quit
```

```
[R2]int loopback 0
[R2-LoopBack0]ip address 10.0.2.1 24
[R2-LoopBack0]quit
```

```
[R3]int loopback 0
[R3-LoopBack0]ip address 10.0.3.1 24
[R3-LoopBack0]quit
```

3. Configure OSPF area 0 on R1, R2, and R3.

```
[R1]ospf 1 router-id 10.0.1.1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]network 10.0.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]network 192.168.2.254 0.0.0.0
[R1-ospf-1-area-0.0.0.0]network 192.168.3.254 0.0.0.0
[R1-ospf-1-area-0.0.0.0]quit
[R1-ospf-1]quit
[R1]
```

```
[R2]ospf 1 router-id 10.0.2.1
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0]network 10.0.2.1 0.0.0.0
[R2-ospf-1-area-0.0.0.0]network 192.168.2.1 0.0.0.0
[R2-ospf-1-area-0.0.0.0]quit
[R2-ospf-1]quit
[R2]
```

```
[R3]ospf 1 router-id 10.0.1.1
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0]network 10.0.3.1 0.0.0.0
[R3-ospf-1-area-0.0.0.0]network 192.168.3.1 0.0.0.0
[R3-ospf-1-area-0.0.0.0]quit
[R3-ospf-1]quit
[R3]
```

4. Check the OSPF neighbor relationship on R1.

```
<R1>display ospf peer
```

```
OSPF Process 1 with Router ID 10.0.1.1
Neighbors
```

```
Area 0.0.0.0 interface 192.168.2.254(GigabitEthernet0/0/1.2)'s neighbors
Router ID: 10.0.2.1      Address: 192.168.2.1
State: Full  Mode:Nbr is Master  Priority: 1
DR: 192.168.2.1  BDR: 192.168.2.254  MTU: 0
Dead timer due in 40 sec
Retrans timer interval: 5
Neighbor is up for 00:08:33
```



```
Authentication Sequence: [ 0 ]
```

```
Neighbors
```

```
Area 0.0.0.0 interface 192.168.3.254(GigabitEthernet0/0/1.3)'s neighbors
```

```
Router ID: 10.0.3.1      Address: 192.168.3.1
```

```
State: Full  Mode:Nbr is Master  Priority: 1
```

```
DR: 192.168.3.254  BDR: 192.168.3.1  MTU: 0
```

```
Dead timer due in 32  sec
```

```
Retrans timer interval: 5
```

```
Neighbor is up for 00:00:13
```

```
Authentication Sequence: [ 0 ]
```

```
<R1>
```

5. Check the OSPF routes on R2.

```
<R2>display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.2.1
```

```
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.2.1/32	0	Stub	10.0.2.1	10.0.2.1	0.0.0.0
192.168.2.0/24	1	Transit	192.168.2.1	10.0.2.1	0.0.0.0
10.0.1.1/32	1	Stub	192.168.2.254	10.0.1.1	0.0.0.0
10.0.3.1/32	2	Stub	192.168.2.254	10.0.3.1	0.0.0.0
192.168.3.0/24	2	Transit	192.168.2.254	10.0.1.1	0.0.0.0

```
Total Nets: 5
```

```
Intra Area: 5  Inter Area: 0  ASE: 0  NSSA: 0
```

6. Ping loopback0 of R3 from loopback0 of R2.

```
<R2>ping -a 10.0.2.1 10.0.3.1
```

```
PING 10.0.3.1: 56  data bytes, press CTRL_C to break
```

```
Reply from 10.0.3.1: bytes=56 Sequence=1 ttl=254 time=90 ms
```

```
Reply from 10.0.3.1: bytes=56 Sequence=2 ttl=254 time=90 ms
```

```
Reply from 10.0.3.1: bytes=56 Sequence=3 ttl=254 time=90 ms
```

```
Reply from 10.0.3.1: bytes=56 Sequence=4 ttl=254 time=80 ms
```

```
Reply from 10.0.3.1: bytes=56 Sequence=5 ttl=254 time=100 ms
```

```
--- 10.0.3.1 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 80/90/100 ms
```

```
<R2>
```



4 Verification

Omitted.

5 Quiz

1. What configurations must be performed on the switch interface connected to the router when inter-VLAN communication is implemented through sub-interfaces?
2. What is the function of the **arp broadcast enable** command?
3. What is the function of the **dot1q termination vid** command?